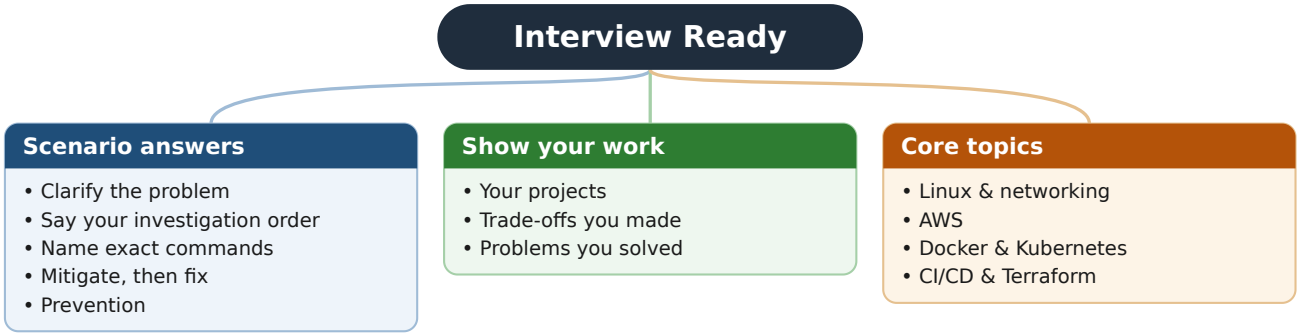
18. Interview Questions (Quick Answers)

In this section: 18.1 Linux & networking · 18.2 Git & CI/CD · 18.3 AWS · 18.4 Docker & Kubernetes · 18.5 Terraform, monitoring, GitOps, platform · 18.6 How to answer scenario questions



18.1 Linux & networking

Question	Short answer
How do you find what is using port 8080?	`ss -tulpn | grep 8080` or `lsof -i :8080`
Disk is full — what do you do?	`df -h` → `du` to find big directories → clean logs/caches/images → check deleted-but-open files (`lsof +L1`) → add monitoring
Difference between a process and a service?	A process is a running program; a service is a process managed by systemd to run in the background
What does `chmod 755` mean?	Owner rwx, group r-x, others r-x
What is load average?	Average number of processes running or waiting for CPU over 1/5/15 minutes; compare with the CPU core count
What happens when you type a URL?	DNS → TCP handshake → TLS handshake → HTTP request → response → render
TCP vs UDP?	TCP reliable and ordered with a handshake; UDP fast with no guarantees
Private vs public subnet?	Public has a route to an Internet Gateway; private does not

18.2 Git & CI/CD

Question	Short answer
Merge vs rebase?	Merge keeps history with a merge commit; rebase rewrites commits into a straight line
How do you undo a pushed commit?	`git revert <hash>` (safe for shared branches)
CI vs CD?	CI builds and tests every change; CD automatically delivers/deploys passing changes

Question	Short answer
How do you store secrets in pipelines?	In the CI secret store or a secrets manager; prefer OIDC short-lived credentials over stored keys
Blue/green vs canary?	Blue/green switches all traffic at once between two environments; canary shifts traffic gradually
A pipeline failed — what do you do?	Find the failing stage, read the first error, reproduce locally, fix the cause, re-run

18.3 AWS

Question	Short answer
Security Group vs NACL?	SG: stateful, instance level, allow only. NACL: stateless, subnet level, allow + deny
IAM user vs role?	User has long-term credentials; role is assumed for temporary credentials
How does a private instance reach the internet?	Route `0.0.0.0/0` to a NAT Gateway in a public subnet
Multi-AZ vs read replica?	Multi-AZ = high availability (standby); read replica = scaling reads
How do you make a web app highly available?	ALB across 2+ AZs, Auto Scaling Group, Multi-AZ database, health checks
How do you control AWS costs?	Budgets/alerts, tags, right-sizing, Savings Plans/Spot, delete unused resources
EC2 is unreachable — what do you check?	State and status checks, public IP, SG, NACL, route table, key/user

18.4 Docker & Kubernetes

Question	Short answer
Image vs container?	Image = template; container = running instance of it
CMD vs ENTRYPOINT?	ENTRYPOINT = fixed command; CMD = default arguments, easy to override
How do you reduce image size?	Slim/distroless base, multi-stage builds, `.dockerignore`, fewer layers
Pod vs Deployment?	Pod = running containers; Deployment manages replicas, rolling updates and rollbacks
Liveness vs readiness probe?	Liveness restarts unhealthy containers; readiness controls whether traffic is sent
Pod in CrashLoopBackOff — steps?	`describe` (events, last state) → `logs --previous` → check config/secrets/command/limits
How do Services find pods?	By label selectors; check with `kubectl get endpoints`
ConfigMap vs Secret?	ConfigMap for normal config; Secret for sensitive data (needs encryption + RBAC)
Requests vs limits?	Requests = reserved for scheduling; limits = maximum allowed

18.5 Terraform, monitoring, GitOps, platform

Question	Short answer
What is Terraform state and why remote?	Record of managed resources; remote (S3 + locking) so teams share it safely
`plan` vs `apply`?	Plan previews changes; apply makes them
What is drift?	Real infrastructure differs from code (manual changes); detect with `plan`
Metrics vs logs vs traces?	Numbers over time / event records / request path across services
SLI vs SLO vs SLA?	What you measure / target / contractual promise
What is GitOps?	Git is the source of truth; an in-cluster agent (Argo CD) pulls and reconciles
What is a golden path?	The supported, pre-built way for developers to do a common task
DevOps vs Platform Engineering?	DevOps is the culture and practices; platform engineering builds self-service tools that make them easy

18.6 How to answer scenario questions

1. Clarify: what exactly is failing, for whom, since when, what changed?
2. Say your investigation order out loud (outside in: DNS → network → load balancer → server → app → database).
3. Name the exact commands or consoles you would use.
4. Mitigate first (rollback/scale), then find the root cause.
5. Finish with prevention: monitoring, alerts, automation, documentation.

REMEMBER

- Behavioral questions: **STAR** = Situation, Task, Action, Result.
- "I don't know, but here is how I would find out" is a good answer.